

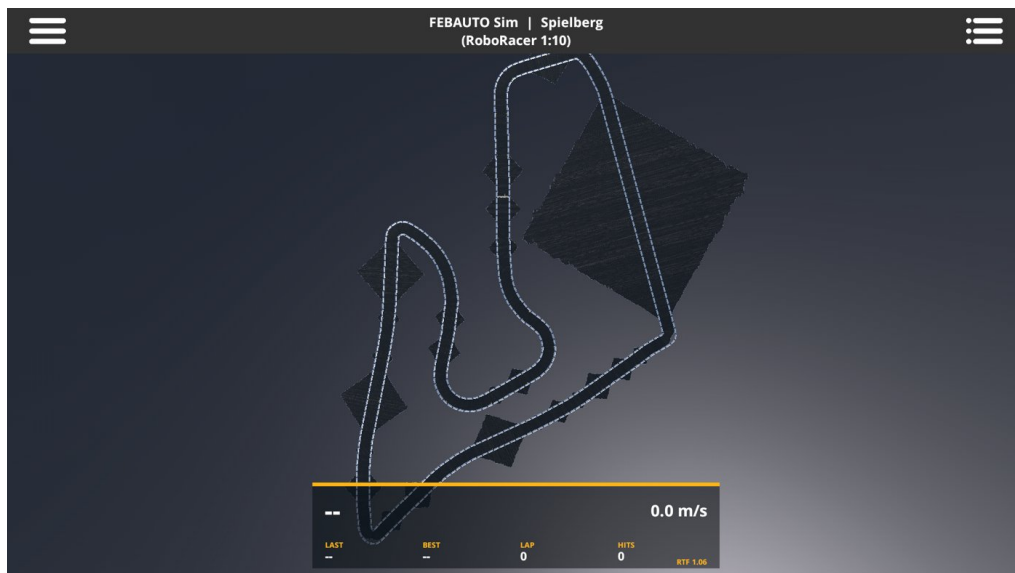


Autonomous Racing Hardware

Options, costs and benefits

Formula Electric at Berkeley, Autonomous Team

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The FEBAUTO simulator running the Red Bull Ring (Spielberg) at 1:10 scale.

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1 Executive summary

The autonomous team has built a complete software racing platform in the last month: a native simulator of the RoboRacer 1:10 race car, a containerised ROS 2 devkit, a starter driving stack, a competition harness, a leaderboard and two manuals. It onboards a new member in an hour and runs a full season (qualification, time-attack, head-to-head final) with no hardware. Its one limit is that the code never meets a real car.

This report lays out five options for closing that gap, from staying software-only to a 1:5-scale vehicle, with a full bill of materials for each, and recommends **Plan B: one RoboRacer-standard 1:10 car with a Hokuyo lidar and a lab track, about \$3,200**. One car is the whole ask: head-to-head racing stays in the simulator, where it already works, because the skills built here are meant to transfer to the FSAE car, and a second small car does not serve that. The two options the committee asked about are analysed in sections 4.4 and 4.5. Converting the team’s Tamiya chassis is not feasible for reasons of eligibility and community rather than cost. A 1:5-scale car becomes affordable once the team’s own RoboSense M1 lidar and the Alameda test track are counted, but it can only be run off campus, which rules it out as the car members learn on day to day; it is the right second step once the 1:10 programme is running. The depth camera and the M1 lidar the team already owns are priced at zero wherever they are used.

Plan	What	Cost (USD)	Verdict
0	Simulator only (status quo)	0	keep as the base of everything below
A	One 1:10 car, budget sensors and compute	1,700	viable; upgrade path to B
B	One 1:10 car, RoboRacer reference build	3,200	recommended
C	Convert the team’s Tamiya 1:10 chassis	1,300	not feasible for competition; poor value
D	1:5-scale vehicle with the team’s RoboSense M1 lidar, tested at Alameda	3,500	viable off campus only; a later step, not the first car

Table 1: The options at a glance. All costs include compute, lidar, power, mounting and the lab track kit; prices are September 2026 list prices, rounded, to be verified at order time.

2 What exists today: the FEBAUTO simulator platform

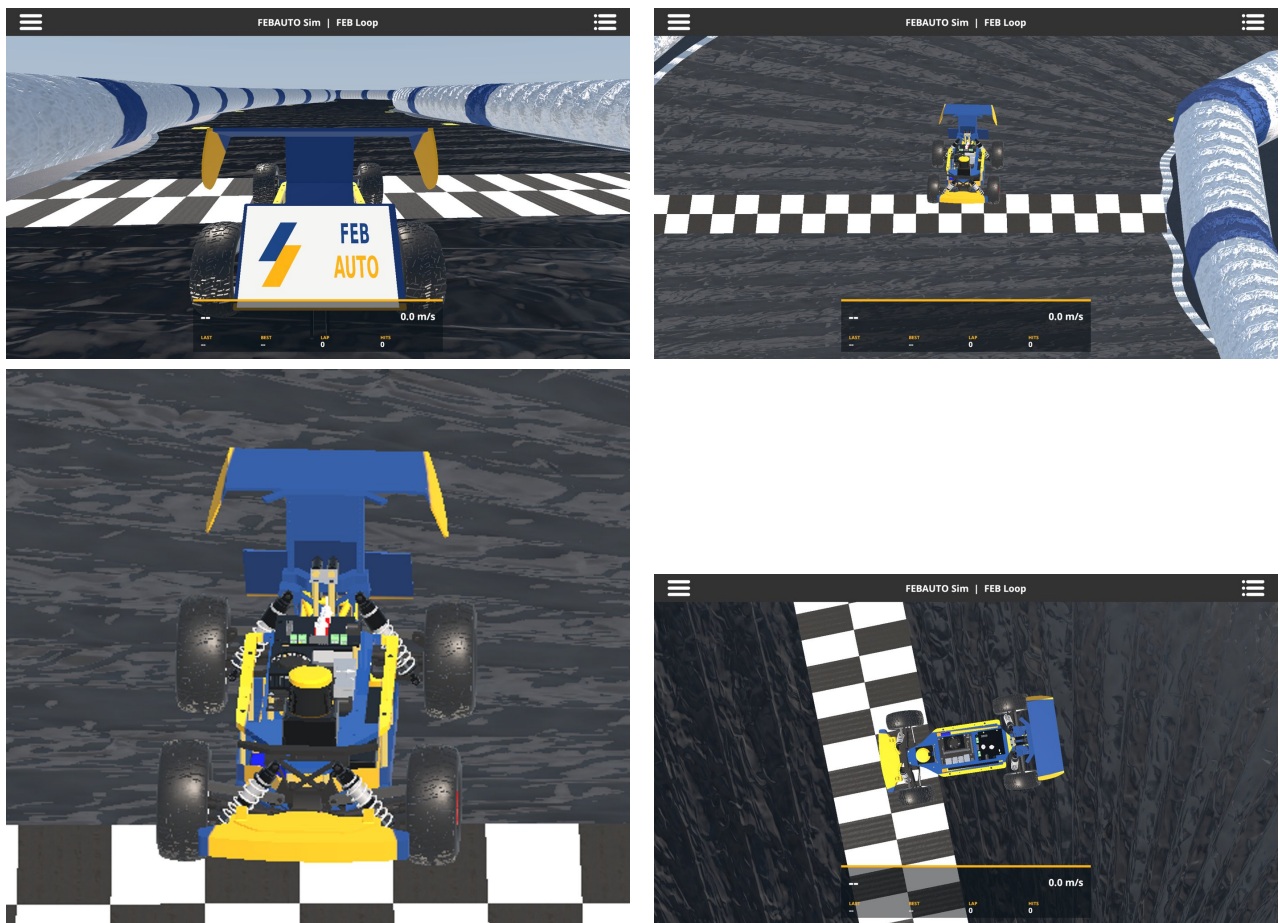


Figure 1: The RoboRacer car in the FEBAUTO simulator: chase view with the race HUD, front view, and two overhead views. The vehicle model is the league’s digital twin, unchanged; the livery and dressing are cosmetic.

2.1 The platform

Piece	What it does
Simulator	Native Mac, Windows and Linux app, a fork of the AutoDRIVE simulator. Tracks load from folders at run time (walls of 33 cm air duct, or FSAE-style cones). Up to four cars. Visual and bare looks. Race HUD, ghost lap, real-time factor readout.
Devkit	Docker image with ROS 2 Humble and the simulator bridge; the member’s code runs inside it; Foxglove for live visualisation on any laptop with nothing else installed.
Starter kit	A commented reactive driver that laps any track, tuning file, system-identification tool, and a learning path from tuning to mapping, racelines and model-predictive control.
House racer	A conservative opponent car for practising overtakes. Starts only when the member’s own stack starts.
Harness	Runs every submitted image on a secret track under league rules, records bags, audits banned topics, writes results, seeds and runs knockout brackets.
Site	Practice boards per track, qualification progress per team, event standings and brackets.
Manuals	Competitor manual (install to leaderboard entry) and organiser manual (tracks, events, race day, releases).

2.2 Tracks

Tracks are folders of a PNG and a YAML file, so a new one takes minutes and needs no rebuild. The team ships four: the FEB Loop, the same loop with FSAE cones instead of walls, the RoboRacer league's Porto practice track, and a custom circuit with a hairpin, esses and a chicane. Real circuits from the RoboRacer track database import directly: the Red Bull Ring at 1:10 scale, 342 m long, is shown on the cover and below.

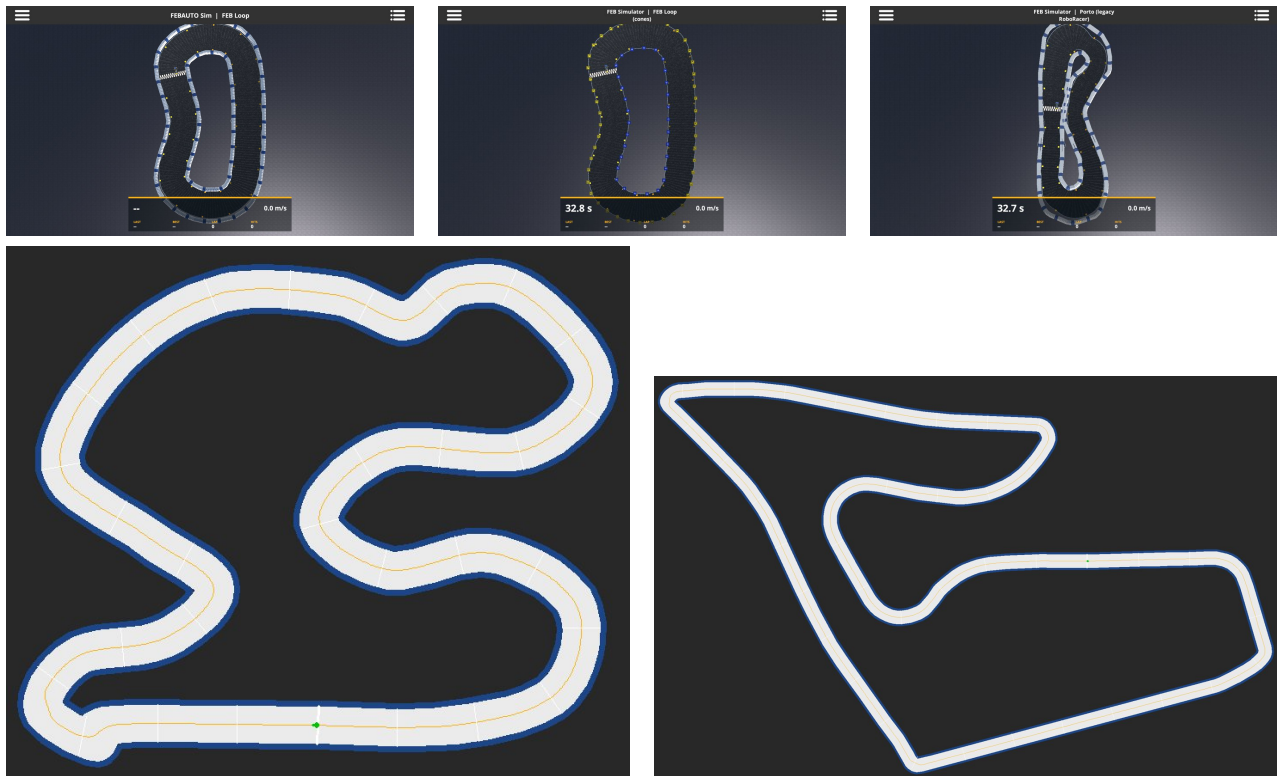


Figure 2: Top: the FEB Loop, its cone version, and Porto, in the simulator. Bottom: the generated maps of the custom FEB Circuit (138 m) and of Spielberg at 1:10 (342 m), with walls, centreline and checkpoints.

2.3 Onboarding and the season

A new member clones one repository, runs one setup command and drives manually. Copying the starter kit and running one more command puts an autonomous car on track; the first leaderboard entry follows the same afternoon. To qualify for the first competition a team completes ten laps with at most one collision on every practice track. Qualified teams race a time-attack on a secret track, and the top seeds meet in a two-car knockout final.

3 Why hardware, and why in the off-season

The FSAE car is not available for autonomous work for most of the year: it is being designed, built or raced by the rest of the team, and it is never available to an untested stack. A 1:10 car is where the autonomous team learns and validates in the meantime, on a table in the lab, with skills that carry straight to the full-size car.

- **People need to see a physical effect.** A lap time on a leaderboard keeps a member for a semester; a car that turns because of code they wrote keeps them for years. The motivation to debug a controller at midnight comes from watching it fail on a real car.

- **Real dynamics cannot be simulated away.** Tyre grip that varies with the floor, a steering servo with backlash, a battery that sags as it drains, a lidar that sees chair legs and reflections: the simulator models none of these faithfully, and the FSAE car has all of them. Members who have tuned a controller against them at 1:10 arrive at the FSAE car knowing what to expect.
- **Sim-to-real training.** The simulator’s car is the digital twin of the RoboRacer reference vehicle. The team’s method becomes: develop and race in the simulator, transfer to the 1:10 car, measure the gap, close it, then carry the same pipeline to the FSAE car. Without the car the first transfer never happens and the method is untested.
- **Perception on real sensors.** Cone detection and colour classification with the camera, lidar clustering, camera plus lidar fusion, localisation against a map: all of it can be built in the simulator, but only real sensors show the lighting, noise, occlusion and latency that decide whether it works. That is the driverless FSAE problem at small scale.
- **Educating new members in real hardware.** Wiring, power, safety procedures, motor controllers, sensor mounting and calibration, ROS 2 on an embedded computer: the 1:10 car teaches every one of these at low cost and low risk before anyone touches the FSAE vehicle.
- **Competitions and recruiting.** RoboRacer runs physical races at ICRA, IROS and IFAC every year on the rules our league already uses, and a car on a table is what fills a room at the first meeting of the semester.

4 The options

Every plan includes compute, lidar, power, mounting and the track kit of section 5. Where a part is available in more than one grade, the table names the one priced.

4.1 Plan 0: simulator only

Cost \$0. Everything in section 2 continues. No sim-to-real work, no physical competition, no perception on real sensors. Kept as the base layer of every other plan, not as an answer on its own.

4.2 Plan A: one 1:10 car, budget grade

Item	USD	Note
Traxxas Slash 4x4 VXL (TRA68086)	500	the RoboRacer-eligible chassis; includes brushless motor and stock ESC
VESC-compatible controller, Flipsky mini (75100 or 6.7)	130	replaces the stock ESC; wheel odometry and IMU
Compute: Raspberry Pi 5, 8 GB, cooler, SD card	120	enough for lidar-only reactive driving and ROS 2
2D lidar: Slamtec RPLidar A2M12	320	12 m, 12 Hz, 0.25°; competition-legal
Two 3S 5000 mAh LiPo, charger, safety bag	190	
Power: buck converter, XT60 splitter, fuse	30	instead of the RoboRacer powerboard
Plates, standoffs, fasteners, USB hub, cables	90	plates laser-cut on campus from the published drawings
Spares: tyres, pinion, bumper	60	
Track kit (section 5)	240	
Total	1,680	about \$1,400 without the track kit

Benefits: the cheapest real car that runs the same ROS 2 stack as the simulator and is eligible for RoboRacer events. **Limits:** the Pi cannot run camera perception at useful rates and the lidar's 12 Hz and range noise cap safe speed at roughly 3 m/s. Both are upgrades, not rebuilds: swap the compute and lidar and Plan A becomes Plan B.

4.3 Plan B: one 1:10 car, RoboRacer reference build (recommended)

Item	USD	Note
Traxxas Slash 4x4 VXL (TRA68086)	500	chassis with brushless motor
VESC 6 MkVI	260	the reference controller
Compute: NVIDIA Jetson Orin Nano developer kit	250	camera perception and learning-based stacks
2D lidar: Hokuyo UST-10LX	1,500	the reference sensor; 40 Hz, 270°, 10 m, the same as the simulator's
Two 3S 5000 mAh LiPo, charger, safety bag	190	
RoboRacer powerboard	100	clean rails for Jetson, lidar and VESC; safe shut-down
Plates, standoffs, fasteners, USB hub, cables	90	
Spares: tyres, pinion, bumper	60	
Depth camera	0	the team's Intel RealSense, already owned
Track kit (section 5)	240	
Total	3,190	about \$2,950 without the track kit

Benefits: the exact car the league, the build guide and our simulator model; camera plus lidar fusion is possible on day one; nothing to upgrade later. **Limits:** the lidar is half the cost and the most fragile part; a bumper and a speed limit for new stacks are part of the plan.

Why not the team's RoboSense M1 on this car. The M1 is a 3D solid-state lidar built for full-size vehicles: about 0.7 kg against the Hokuyo's 130 g, a 120° forward field of view against 270°, and a near-field blind zone of roughly half a metre in a corridor with under a metre of free space either side. It would dominate the car's mass, see nothing beside it, and is outside the league's sensor set. It is the right sensor for the 1:5 option below and for the FSAE car; on the 1:10 car the 2D lidar is the right tool.

4.4 Plan C: convert the team's Tamiya 1:10 chassis

The team owns a motorless Tamiya touring-car chassis. Priced as a full conversion:

Item	USD	Note
Tamiya chassis	0	owned
Brushless motor and ESC or VESC, steering servo	200	the kit has neither
Compute, lidar, power, cables as Plan A	760	
Custom sensor deck and mounts (design, prints, iterations)	80	no published drawings exist for this chassis
Track kit	240	
Total	1,280	saves about \$400 against Plan A

Why it is not feasible as the team's car:

- **Not competition-eligible.** RoboRacer rules require a Traxxas 1:10 chassis (codes TRA74054, TRA6804R, TRA68086). A Tamiya cannot enter any physical event.
- **Not the simulated car.** Wheelbase, track width, tyre model and steering geometry differ from the digital twin, so simulator tuning does not transfer, which removes the main reason to have a car.
- **No community.** Every RoboRacer document, drawing, VESC profile and forum answer assumes the Slash. On the Tamiya the team solves mounting, power and drivetrain questions alone. That is a semester of mechanical work that teaches no autonomy.
- **Weak platform.** A touring car has 5 mm of ground clearance and no room for a lidar, a computer and a battery. The Slash was chosen by the community because it carries them.

The chassis is still useful as a bench mule for wiring and sensor mounting practice, at no cost.

4.5 Plan D: a 1:5-scale vehicle with the M1 lidar, tested at Alameda

The committee asked about a 1:4 car. No manufacturer makes a 1:4 road car; the nearest production platforms are 1:5 short-course trucks (Losi 5IVE-T, Arrma Kraton 8S) and 1:6 monster trucks (Traxxas X-Maxx). Two things the team already has change this option's cost: the RoboSense M1 lidar, which is the right sensor at this scale, and access to a test track in Alameda, which removes the need to build a track.

Item	USD	Note
1:5 electric chassis (8S class)	1,300 to 1,900	8 to 12 kg, 50 km/h and up
8S battery pair and charger	500	
High-current controller (VESC 75V/300A class)	350	
Compute: NVIDIA Jetson Orin NX class	700	the M1's 75,000 points per frame need more than an Orin Nano
Lidar: RoboSense M1	0	owned
Depth camera: Intel RealSense	0	owned
Custom deck, roll protection, kill switch, remote stop	450	nothing published to copy
Track: Alameda test track	0	owned access; transport and session costs not included
Total	3,300 to 3,900	

What it would give the team. A vehicle with a 3D lidar and a camera at a size where the FSAE perception stack runs unchanged: the same sensor, the same point-cloud pipeline, the same cone detection, on a car that can be built and crashed without touching the FSAE vehicle. For the driverless FSAE goal this is closer to the real thing than any 1:10 car.

Why it is not the first car, stated plainly:

- **It cannot be run on campus.** A 10 kg vehicle at 50 km/h needs a closed outdoor area, a kill switch and a marshal. Every test is a trip to Alameda with the car, a generator or charged packs, and a tow of tools: a session, not an afternoon in the lab. Iteration drops from ten runs a day to one or two a week, and the weather has a vote. New members cannot be handed this car to learn on.
- **Safety.** About 1 kJ of kinetic energy, thirty times the 1:10 car. Untested stacks and first-semester members do not belong near it.

- **No community and no competition.** No autonomous racing series exists at this scale. Mounts, power and motor control are solved alone.
- **Not the simulated car.** Nothing built in the simulator transfers until a 1:5 vehicle model and an M1 sensor model exist in it; both are on the simulator’s future list, and neither is small.

Verdict: viable and worth planning for as the perception mule that sits between the 1:10 programme and the FSAE car, once there is a 1:10 car for members to learn on and a stack worth putting on it. Not the first purchase.

5 The track kit

Item	USD	Note
Flexible aluminium duct, 12–13 inch (33 cm) diameter, 4 lengths of 25 ft (about 30 m total)	130	the league’s walls; folds flat to store; joins with tape
FSAE-style cones, 3D-printed at 1:10 (100 pieces)	15	32 mm tall, blue and yellow filament, about 10 g each; printed on the team’s printers
Floor tape, start-line gate (photo interrupter), storage tub	95	
Total	240	

Thirty metres of duct builds a 12 m by 6 m loop with a hairpin, the same geometry as the FEB Loop in the simulator, in any lab or corridor on campus, laid and packed away in ten minutes. That is the point of it: the Alameda test track exists and is the venue for anything larger, but it is off campus, session-based and weather-bound, and the 1:10 programme has to run every day. The cones give the cone version of the same track for camera work, and scale exactly like the simulator’s cone tracks.

6 Sensor fusion and the driverless connection

The team already owns an Intel RealSense depth camera. With a 2D lidar it gives the sensor pair used by Formula Student driverless cars at small scale: the camera classifies cones by colour and the lidar gives precise range and bearing; fusing them yields a cone map, then a track boundary, then a raceline. The FEBAUTO simulator publishes both a front camera image and a lidar scan on the same topics the real car will use, and its cone tracks exist for exactly this pipeline. Plan B supports this on day one; Plan A’s compute is enough for lidar-only driving and can be upgraded to a Jetson for camera work later.

7 Racing: one on one, in the simulator

The head-to-head final is the season’s finish and it stays in the simulator: two cars on one track, each team’s stack in its own container, contact costing both cars a penalty and a respawn, finishing order by laps then time. The simulator already does this, so no second physical car is asked for. The figures show a race on Spielberg and on the FEB Loop between the starter kit and the house racer.

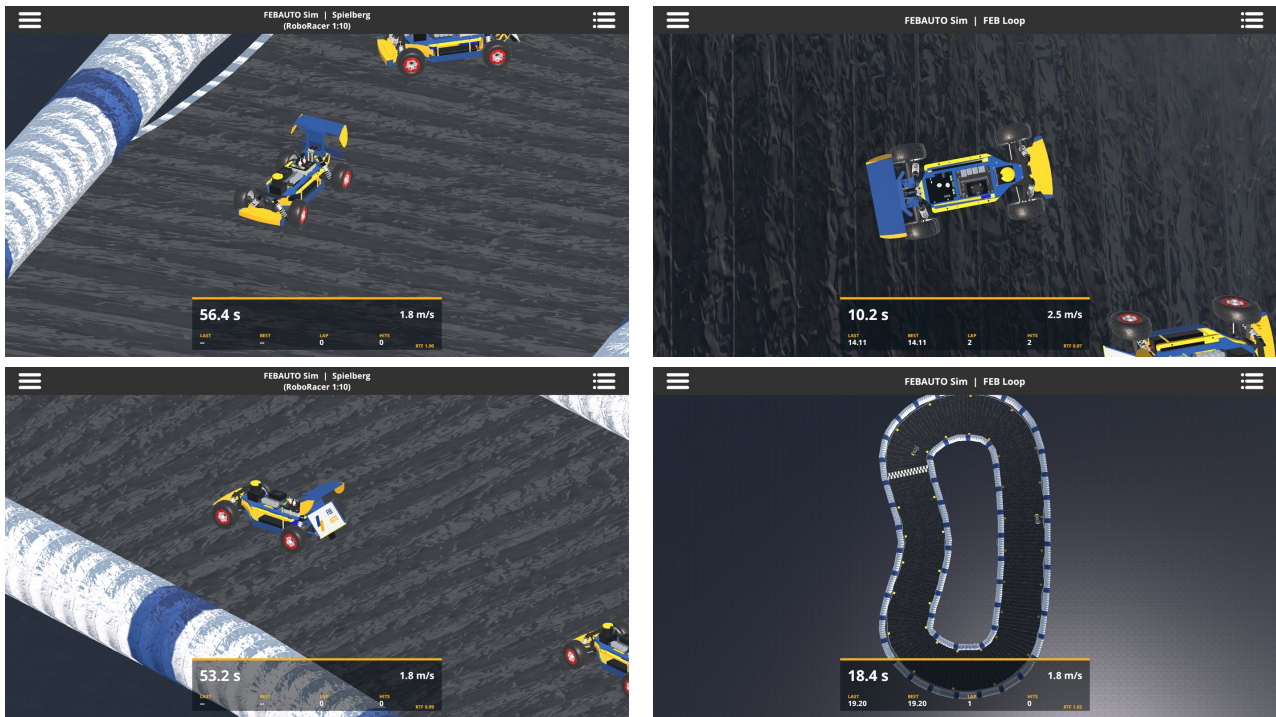


Figure 3: One-on-one AI racing. Top: on Spielberg, the house racer (top of frame, left) closing on the member’s car, and a moment earlier as it enters the frame (right). Bottom: on the FEB Loop, both cars in the same corner from above (left) and the whole track with both cars (right). Each car runs an unchanged competitor image behind the harness proxy; contact costs both cars a penalty and a respawn.

8 Recommendation and timeline

Fund Plan B (\$3,190 plus 10 % contingency, \$3,500). Order the lidar first; its lead time is the longest. Revisit Plan D, the 1:5 car with the M1, once the 1:10 programme has run a season and the simulator carries a 1:5 vehicle and M1 sensor model.

When	Milestone
Order + 2 weeks	Car assembled from the RoboRacer build guide; manual drive; sensors on ROS 2; track laid
+ 4 weeks	Starter kit laps the lab track; simulator-to-car gap measured and written up
+ 8 weeks	Qualified teams run their stacks on the car; first physical time-attack; camera and lidar cone perception on the cone track
Following semester	Entry to a RoboRacer physical event; perception pipeline carried to the FSAE car

9 Risks

- **Crashes.** Bumper, soft duct walls, a speed limit for new stacks; spares budgeted. The lidar is the fragile part and sits behind the bumper line.
- **Batteries.** LiPo bag, supervised charging, a written procedure before the first charge.
- **Supply.** Hokuyo lead times vary; every other part is stocked by hobby and electronics retailers.
- **Access.** The car lives with the autonomous lead; access follows the simulator’s qualification rule, so an untested stack never has the car as its first target.

Platform: <https://github.com/Pranman1/feb-racing> Leaderboard: <https://pranman1.github.io/feb-racing/>

RoboRacer build guide and bill of materials: <https://f1tenth.readthedocs.io>

RoboRacer rules: https://github.com/f1tenth/roboracer_rules Track database: https://github.com/f1tenth/f1tenth_racetracks